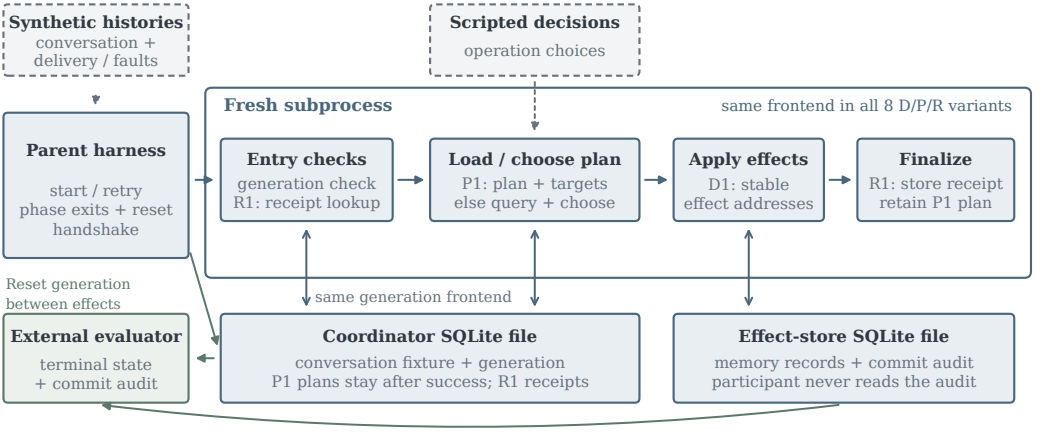
Factorial memory recovery with real process failures



Exit phases: after prepare / first put / all effects / finalize

Abrupt os._exit and committed-effect exceptions; separate storage commit boundaries

executed algorithm, storage or process control — controlled synthetic input or decision source